

[Request] Which is more and by what magnitude more energy efficient : asking chatgpt or similar LLM in a server farm or asking people on Reddit?

Posted by u/nasazh • 0 points • 2 comments

I've had a question about specific functionality of ZigBee devices, and asked about it on a subreddit. Couple of people answered and I got an answer, fast, a bit different than I expected, but really good.

Then I tried same question verbatim on chatgpt. It firstly gave me some bloated wall of text, and after I've logged in and enabled reasoning and search it gave me quite an ok answer with on par information, but with different focuses which I've would have probably missed if I didn't know the answer already.

What are electric energy bill in first option and second? Like server for reddit to facilitate 3 people exchanging couple messages plus 3 computers plus networking and server for LLM plus 1 computer plus networking.

Approximately of course.

Comments

u/Memer_Plus • 1 points • 2025-02-18 01:48 UTC

According to [this](<https://cacm.acm.org/blogcacm/the-energy-footprint-of-humans-and-large-language-models/>), an LLM query takes up 4 J of energy per token. Your ZigBee post has 645 characters, so with a ratio of 1 token to 4 characters, it takes up 162 tokens, and thus 648 J of energy, for your question in an llm.

Assuming that Reddit has similar energy consumption to other social media platforms, you would use 10,800 J of energy every day. I got it from [here] (https://www.reddit.com/r/askscience/comments/b4sk2f/how_much_energy_does_an_average_post_on_social/). Assuming that you are an average Reddit user, you used 1,762 characters in the last 24 hours. So for every character you use, you use up 6.13 J *per character*. Therefore, here, you use 24.5 J per token. You use 6 times more energy with a Reddit post than an AI query.

###General Discussion Thread

This is a [Request] post. If you would like to submit a comment that does not either attempt to answer the question, ask for clarification, or explain why it would be infeasible to answer, you **must** post your comment as a reply to this one. Top level (directly replying to the OP) comments that do not do one of those things will be removed.

I am a bot, and this action was performed automatically. Please [contact the moderators of this subreddit](/message/compose/?to=/r/theydidthemath) if you have any questions or concerns.